

The extra learning assumption

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September 11, 2023

Making it constructive

The extra is the following:

In this situation, the UDT prior (the infinite $\mathbb{P}$ that decides policies) is such that, for every n , UDT eventually acts as if using (the infinite extrapolation of) $\mathbb{Q}_n$ instead of $\mathbb{P}$.

In fact, **this already sometimes happens contingently**, without our intervening in any way. (That is, just using as $\mathbb{P}$ the infinite extrapolation of an early $\mathbb{Q}_k$, as we usually do.) Indeed, consider for example a tree where Ω , through logical coins, just incentivizes satisfying that assumption. Maybe Ω has an infinite sequence of coins of the form "in the rounds between $f(n)$ and $f(n+1)$, the agent's actions are the same as those it would take if using the knowledge from $\mathbb{Q}_n$ " (and say, between those coins, there are some independent decision problems¹, whose rewards are way lower than those obtained from making the logical coins true). Then, if we use as our frozen prior a late enough $\mathbb{Q}_k$, it will perfectly understand that conditioning on something like $\forall m, n(f(n) \leq m < f(n+1) \rightarrow \mathbf{A}_m = \text{Procedure}(\mathbb{Q}_n))$ increases U a lot (where Procedure is just a fixed description of the way we use a probability distribution to decide actions).

Usually, though, $\mathbb{Q}$ will have good reasons to have opinions on the far-off future. For example, if it knows $\mathbf{A}$ will face infinitely many Counterlogical Muggings (but, of course, doesn't yet know all the logical coins), and has credence $\mathbb{Q}_k(\varphi) = 0.9$, it will have the universal belief that "if $\mathbf{A}$ ever sees a Counterlogical Mugging with φ , and the payoff ratio r is $> 1/9$, it should take it (instead of, for example, thinking longer)".

Since we want to (when extrapolating $\mathbb{Q}_k$ into our infinite frozen prior $\mathbb{P}$) ensure it eventually updates, **we will necessarily have to trash knowledge** of this kind. We can't have our cake (preserve all the knowledge in $\mathbb{Q}$) and eat it (ensure it has these opinions instead of those ones).

A hackish but natural way to ensure this is just "hard-coding" something like "in the rounds between $f(n)$ and $f(n+1)$, the way to maximize U is just using the knowledge from $\mathbb{Q}_n$ ". As mentioned, we'll have to prune some knowledge: $\mathbb{P}$ cannot keep thinking "I should always pay up in this Mugging". One way to implement the hard-coding is as follows:

¹We can think of them as purely empirical problems to avoid undesirable logical correlations that might complicate the picture. But actually, one can think of some logical problems that are intuitively "independent enough" of the logical coins such that $\mathbb{Q}$ would also converge on ignoring the correlations. For example, correctly guessing digits of π (under the hypothesis that they are uniformly pseudorandom, AND that they don't share any secret polynomial-time correlation with our logical coins).

Consider all sentences in $\mathbb{Q}_k$'s support that contain the syntactic shape $\mathbf{A}_m = a$, that is, the syntactic shape that represents actions explicitly, and the one we directly condition on when deciding actions. This doesn't only mean the sentences $\mathbf{A}_m = a$, but also any sentences including them as subformulas, like $\forall \mathcal{Q}, m(\mathbb{Q}_m = \mathcal{Q} \wedge P(\mathcal{Q}) \wedge \mathbf{A}_m = g(\mathcal{Q}) \rightarrow U \geq 3)$. Change all of these sentences so that they only talk about actions before $f(0)$. For example, the above one would look $\forall \mathcal{Q}, m(m < f(0) \rightarrow (\mathbb{Q}_m = \mathcal{Q} \wedge P(\mathcal{Q}) \wedge \mathbf{A}_m = g(\mathcal{Q}) \rightarrow U \geq 3))$. This is a well-defined syntactic operation, that we are performing on the finite probability distribution $\mathbb{Q}_k$. The resulting distribution still fits our basic coherence properties². Notice the resulting distribution might still have opinions beyond $f(0)$. We'll just replace all of those by something arbitrary that ensures, when deciding which policies to enact, the policy of "follow Procedure($\mathbb{Q}_n$)" is always enacted (between $f(n)$ and $f(n+1)$). For example, on top of the knowledge about U granted by the still-present beliefs about $< f(0)$, we can hard-code (as extensions to all those coherent worlds, that is finite conjunctions), with high enough probability, that enacting that policy leads to an additional increase of U by 5, and not doing it always leads to an increase below 4 (or any other arbitrary thing like that).

I know it feels bad to settle for something so arbitrary, but we already knew that, necessarily, we'll be trashing arbitrarily much knowledge by $\mathbb{Q}_k$ about the future. It's true in some situations we could get away with leaving some of that knowledge untouched. For example, $\mathbb{Q}_k$ might have mostly inconsequential opinions about the far-off decision tree, that are consistent with our hard-coded implementation. Or even, that imply it with high probability (as in our first example above). But what would we gain from preserving that knowledge? Nothing, since after $f(n)$ we'll be using $\mathbb{Q}_n$, that already has all such knowledge (or actually, improved later versions of it). That is, we've changed our single-time extrapolation in a hacky way to ensure our assumption is met, but $\mathbb{Q}_n$ keeps of course functioning as usual.

Refinements of this construction

Maybe we feel bad about this because it's just too close to Policy Selection. We've basically successfully shoe-horned Policy Selection into the prior. But I feel like that's the spirit of what we'll be doing (with more or less aesthetic grace) if we demand learning. It is true, thought, there are probably other similar but subtler procedures with more aesthetic grace, or that seem more cognitively natural, or to include more nuance.

Indeed, consider the following refinement:

We already know that, if we don't change / nudge the prior in any way towards updating, it might never update. What if we do implement changes so that it forcibly updates, but nonetheless we **employ its knowledge about the future to softly guide this implementation** in finite times? For each round or situation (or class of situations, partitioned in some way), say we check $\mathbb{Q}_k$'s complete opinion (about the future) on "how bad it would be to update (use Procedure($\mathbb{Q}_n$)), as opposed to carrying out the policies that I consider best". In some cases this will be "not bad at all", since it already recommended being that updateful. In others, it will be "super bad", since it's pretty sure that it can make a lot of money in this Counterlogical Mugging, and that will lose it. In others, it will be "meh", etc.

Then, when we have to perform the brain surgery to ensure the prior eventually updates, we can use these estimates (the amount of U predicted to be lost) to make some sentences or classes update

²I strongly think that's the case, but I should formalize the proof.

faster or slower. For example, if $\mathbb{Q}_k$ thinks updating on this one situation is very bad, we can be gentle, and let that take a longer time to start updating, and then update only slowly. Nonetheless, we must be firm, and ensure that the update eventually comes, even if we have to drag $\mathbb{Q}_k$ there kicking and screaming.

We can also expand this by, instead of just "eventually using the $\mathbb{Q}_n$ knowledge", enforcing (inside our description of Procedure) that "eventually (when enough updates have been carried out) **re-freezing the whole prior for a while** (brain surgery to ensure eventual updating included), and doing all of this again". That is, we periodically "reassess" how fast we should update. We could also periodically reassess how fast we should reassess about how fast we should update, etc. Any finite meta-level is implementable. I can also imagine implementing some approximations to ω levels, like "in day n^n you for the first time reassess the meta- n -level, and from there on reassess every n^n days (until the meta- $n+1$ -level kicks in, and then it will take the reigns and start deciding)".

On a similar note, instead of just looking at current probabilities to assess "how bad does it seem to be updateful about this", we can look at past market history to check "how stable that belief has been", and make this stability also come into the decision of how fast to update (a stable belief should count for more).

This all sounds nice, but I think they're just fancier implementations of Policy Selection (with more bells and whistles), while remaining compatible with the epistemic LI / "beliefs looking like beliefs" picture (except, maybe, for the fact that we have to enforce brain surgeries).

Universal instantiation

I think your worries about universal instantiation can probably be put to rest.

The way in which universal instantiation is fundamental to our construction is in extrapolation, to ensure that $\mathbb{Q}_k$'s small beliefs can later be translated into large beliefs representing them. And that's of course preserved by constructions like the above (together with other kinds of coherence). Importantly, we don't need "our LI to be Gaifman Inductive", or anything like that. We just need the infinite extrapolation (that we syntactically carry out once and for all) to be Gaifman Inductive, so to speak. What we need is that, if $\mathbb{Q}_k$ believes some general statement, this is translated into all particularizations in our prior. And we are able to get this very easily. We don't need "all true Π_1 sentences are eventually learned", since we're not talking about actual truth: just about the fit between small and large beliefs.

Maybe your worry is that, even if that works, the LI we are using is still, of course, not Gaifman Inductive. And so our agent might never learn some facts crucial to obtaining high reward. This might be the case, I don't know how it will pan out when we try and prove things about reward (although I suspect like if we use time-penalized U it won't be a problem). But in any event it'd be a problem with our tools for logical learning (we haven't been able to make LIs Gaifman Inductive), not with our construction of UDT.

The LIC

I'm confused about what you mean by losing the LIC (when we perform a construction like the above).

The $\mathbb{Q}$ that we keep running (and our agent consults) will of course always satisfy the LIC (we are not modifying it in any way, other than by making it propositionally coherent, universally instantiated, etc., which fit nicely in the LIC framework).

And the infinite extrapolated $\mathbb{P}$... well, it's a single probability, so it doesn't make sense to talk about the LIC. Maybe you worry that if we re-freeze infinitely often, that sequence won't satisfy LIC. But I think it does (proof coming soon).

Maybe there are other subtler worries.

The infinite power of extrapolation

I don't have time to write this out now, but basically I had a worry that implementing Procedure() wouldn't be the same as our actually doing the whole infinite extrapolation. But thanks to some nice properties of the infinite extrapolation, it actually is the same!

Okay, but what do we prove?

This thing about "the extra assumption" has been very interesting. But then, what non-obvious results about performance do we want to prove (from that and other assumptions)? That's addressed in my last document "Impossibility results", and I'll work more on that soon :)